

Cambridge IGCSE™

PHYSICS**0625/62**

Paper 6 Alternative to Practical

October/November 2025

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require *n* responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards *n*.
- Incorrect responses should not be awarded credit but will still count towards *n*.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first *n* responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	arithmetic error
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	response has not answered question

Annotation	Meaning
	rounding error
 or 	point has been noted, but no credit has been given or blank page seen
	error in number of significant figures
	transcription error
	response is too vague or there is insufficient detail in response
	answer outside the tolerance of the mark scheme
	used to highlight parts of an extended response
	used to highlight parts of an extended response
	mandatory mark not awarded
	special case
	unclear response
	poor diagram
	power of ten error
	incorrect physics
	incorrect unit

PUBLISHED**Specific Instructions for Marking 0625/Paper 6****Preparation for Marking**

Instructions and handbooks, for markers using RM Assessor 3 can be found at [RM support portal](#).

Marking**M1. Blank pages, additional objects and marking outside the question zone.**

Blank pages will be attached to the first part of Q1 and should be annotated with SEEN on all scripts.

Annotate any blank Additional Objects with SEEN.

Link any other additional objects to the question or questions applicable.

Examiners must ensure that they view the whole exam paper for each candidate. This will sometimes mean scrolling through a large zone to ensure that no working relevant to either the current or any other question is missed.

Where a candidate's answer extends beyond the marking zone, examiners must view the whole page (or link to other pages) to annotate and mark the whole answer. To view the whole page, deselect any annotation tool from the mouse, then click in the bottom right-hand corner of the marking zone where "view whole page" appears. For instructions to link to other pages see above.

M2. Use of Annotation tools.

Examiners annotate scripts to explain their reasons for awarding or not awarding marks, noting:

- for **all** questions with **two** or **more** marks, it is **mandatory** to annotate with ticks placed to indicate where each mark is awarded. In a calculation where the final answer (A) mark is awarded all the ticks should be placed near to the final answer.
- annotations and comments must never suggest or imply that a mark has been deducted e.g. –1
- for questions with only one mark maximum, examiners' ticks to show the mark is awarded are not mandatory
- SF and ECF annotations must be used when appropriate (see table of annotation tools)

M3. Acronyms and shorthand in the mark scheme.

acronym/shorthand	explanation
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word/phrase/unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.

Owtte	Or words to that effect.
Ignore	Indicates either an incorrect or irrelevant point which may be disregarded, i.e., <u>not</u> treated as contradictory.
insufficient	an answer not worthy of credit <u>on its own</u> .
CON	An incorrect point which contradicts any correct point and means the mark cannot be scored.
ecf [question part]	Indicates that a candidate using an erroneous value from the stated question part must be given credit here if the erroneous value is used correctly here.
Cao	correct answer only

M4. Miscellaneous

Equations and formulae. Where a C, B or M mark is available for quoting a formula or equation this can be done in any form and in words, symbols or numbers unless the mark scheme specifies otherwise.

Use of ecf. The mark scheme notes where ecf is applicable, in the guidance section of the final answer mark. However, it should be applied for all relevant C marks as well. **Always annotate ecf if applied.** See Science specific Marking point 4 above.

Units.

- A numerically correct final answer without a unit is awarded the final answer (A) mark if the unit is shown correctly in the candidate's working.
- A numerically correct answer with a missing or incorrect unit is not awarded the final answer (A) mark. C (B or M) marks are awarded from the candidate's working.
- Accept units with incorrect use of upper-case and lower-case symbols, e.g. pA for Pa.
- Unless the mark scheme for a specific question part states otherwise, the only permitted derived units are:

unit	permitted derived units
W	J / s or Nm / s
Pa	N / m ²
momentum	Ns or kgm / s
impulse	Ns or kgm / s

J	Nm
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- NB J is **not** permitted as the unit for moments.

Significant Figures.

- Unless otherwise indicated in the mark scheme final answers expressed to two or more significant figures receive the final answer (**A**) mark if the candidate's answer rounds to the mark scheme answer.
- A final answer expressed to one significant figure is only awarded the final answer (**A**) mark where the final answer is exact to one sig. fig. (This applies to all answers, including answers using ecf.)
- A correct numerical answer, quoted with fewer significant figures than required by the mark scheme (even if in the working it has the required number of significant figures), is not awarded the final answer (**A**) marks. **C** (B or M) marks are awarded as appropriate.

Fractions. An answer expressed as a fraction is not a numerically correct final answer unless the fraction is explicitly stated in the mark scheme.

Crossed out work. When only part of an answer is crossed out the crossed-out work must be ignored. However, work which has been **wholly** crossed out and not replaced and can easily be read, should be marked as if it had not been crossed out. Look to see if it has been replaced on a blank page or another part of the same page before attempting to mark the crossed-out work.

Marking diagrams on-screen. Differences in magnification and/or individual computer screen settings can alter the appearance of diagrams. If it is necessary to check line lengths or angles use the ruler and protractor tools provided within RM Assessor 3 to ensure consistency across all examiners.

NR. (# or / key on the keyboard). Use this (instead of giving 0 marks) if the answer space for a question is completely blank or contains no readable words, figures or symbols.

Question	Answer	Marks
1(a)	normal correct (by eye), extended to cross PR <u>and</u> Q labelled	1
1(b)	$\theta = 10^\circ \pm 1^\circ$ <u>and</u> end of line labelled O $MO = 14.0 \pm 0.2$ cm	2
1(c)(i)	line joining M to the crosses, continued to cross PR	1
1(c)(ii)	$a = 1.7 \pm 0.1$ cm $b = 10.3 \pm 0.1$ cm	1
1(d)	r correct answer to 2 significant figures	2
1(e)	statement to match results (expect YES) <u>values from table</u> used to justify statement	2
1(f)	extend the investigation for different values of θ	1
1(g)	reflecting surface of the mirror is at the back/thickness of mirror/small variations in protractor angle have a much larger effect on a and b / thickness of light ray/difficult to locate centre of light ray/experiment not done in a dark room	1

Question	Answer	Marks
2(a)	4.4(0) (V)	1
2(b)(i)	0.135 /0.14 (A) 0.14 (A)	2
2(b)(ii)	3.0(1)	1
2(c)	scales labelled with quantity and unit, correct way around suitable scales (occupying at least $\frac{1}{2}$ the grid) plots correct to $\pm \frac{1}{2}$ small square <u>and</u> precise plots good best-fit line judgement, thin, continuous line	4

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Question	Answer	Marks
2(d)	indication on the graph as to how gradient found <u>and</u> correct method of calculation of gradient i.e. $\Delta y / \Delta x$ at least $\frac{1}{2}$ the distance between the extreme plotted points used ($\Delta I \geq 0.045$)	2
2(e)	any integer from 19–25 inclusive	1

Question	Answer	Marks
3(a)(i)	87.0 (°C)	1
3(a)(ii)	min/minute(s) <u>and</u> 1, 2, 3, 4, 5	1
3(b)(i)	to get the highest temperature reached/to allow the liquid in the thermometer time to expand/to allow the beaker to warm up/to allow the thermometer to reach the temperature of the hot water/to allow the thermometer (time) to respond/to ensure that the temperature has stopped rising	1
3(b)(ii)	read thermometer perpendicular to the scale/at eye level / hold the stopwatch close to the thermometer/thermometer not touching the sides/base of the beaker/stir the water before taking a reading	1
3(c)(i)	11.5 °C	1
3(c)(ii)	$R_1 = 5.75$ / candidate's (c)(i) $\div 2$ °C / min	2
3(c)(iii)	$R_2 = 3.5$ (°C / min) or 0.0583 (°C / s)	1
3(d)	cooling rate decreases (with temperature/time)	1
3(e)(i)	45–55 (°C) inclusive	1
3(e)(ii)	room temperature/21.5 °C	1

Question	Answer	Marks
4	variable MP1 valid variable which may affect h <u>stated</u>	1
	apparatus MP2 ruler/measuring tape <u>and</u> additional apparatus to measure the independent variable (if not height of release or diameter of ball)	1
	method MP3 <u>measure</u> the independent variable <u>and</u> release the ball MP4 <u>measure</u> h <u>and</u> repeat for a new value of independent variable	2
	control variable MP5 any significant variable appropriate to the independent variable	1
	table MP6 columns <u>with units</u> , for independent variable and dependent variable	1
	conclusion MP7 compare readings in the table to see <u>if/how/whether</u> a change in the variable produces a change in h OR plot a (line) graph <u>with axes specified</u>	1